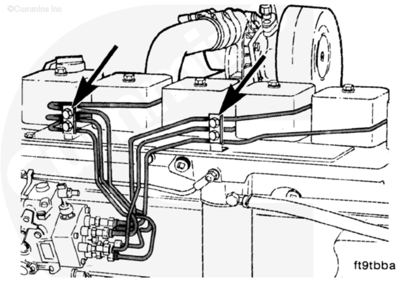
Trainings

General Information

⚠ CAUTION ⚠

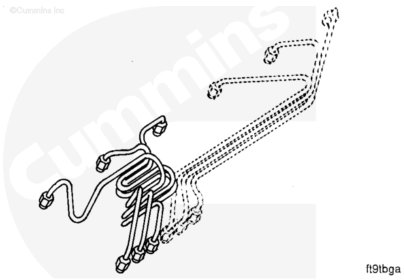
The high-pressure lines must be clamped securely and routed so they do not contact each other or any other components.



The high-pressure fuel lines are designed and manufactured to deliver fuel at injection pressure to the injectors. The high-pressure pulses will cause the lines to expand and contract during the injection cycle.

Shown here are the high-pressure fuel lines for the distributor-type injection pump.

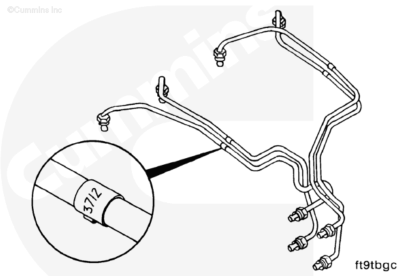
Shown here are the high-pressure fuel lines for the Bosch® in-line injection pump.



Do **not** weld or substitute lines; use **only** the specified part number for the engine.

The length, internal size, and rigidity of the lines are critical to smooth engine operation. An attached metal tag is used to identify each line with a part number.

A metal tag attached to each line identifies the line with a part number.

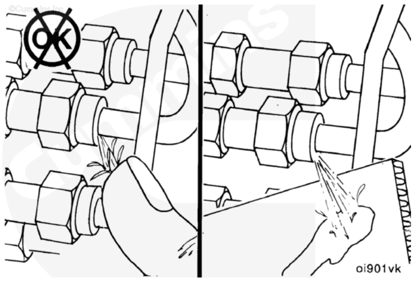


⚠ WARNING ⚠

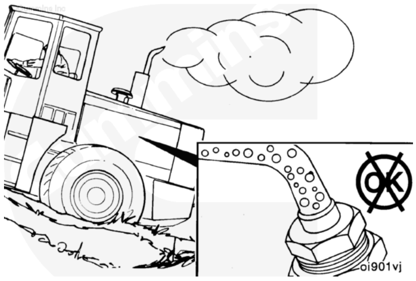
Keep hands and body parts away from the high-pressure fuel lines. Fuel coming from the high-pressure fuel lines is under extreme pressure and can cause serious injury by

penetrating the skin.

Use cardboard to check for cracks and leaks. With the engine running, move the cardboard over the fuel lines, and look for fuel spray on the cardboard. Fuel leaks can cause poor engine performance.



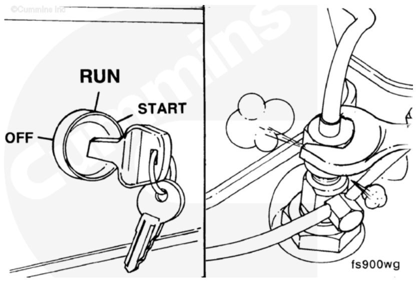
It is normal to have entrapped air in the fuel lines after replacing the pump or the lines. Air in the lines will cause the engine to run rough or produce a fuel knock.



Bleed the air from the high-pressure line at the fitting that connects the injector. Bleed one line at a time until the engine runs smoothly.

Follow the “Prime” step below for the proper procedure.

If the air can **not** be removed, check the pump and supply line for suction leaks. Refer to Procedure 006-003 (/qs3/pubsys2/xml/en/procedures/40/40-006-003-tr.html), Air In Fuel.



Initial Check

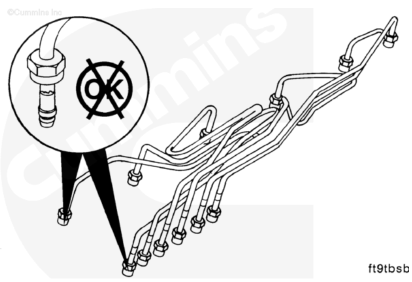
Inspect the lines for cracks, chafes, and leaks. Make sure that the lines are tightened to proper specification.

Pump Side Fitting:

Torque Value: 24 n•m [18 ft-lb]

Fitting at Head:

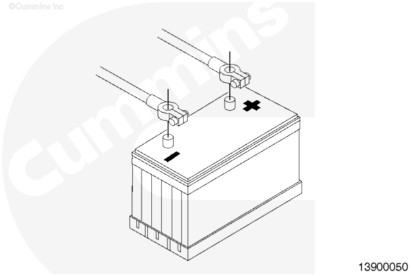
Torque Value: 38 n•m [28 ft-lb]



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

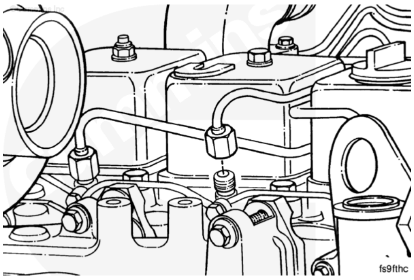


- Disconnect the batteries.

Remove

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

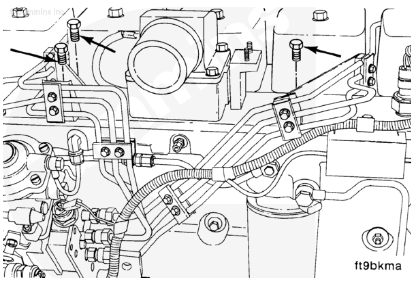


Distributor-Type Pumps

Note : Thoroughly clean the area around the fuel lines before removal.

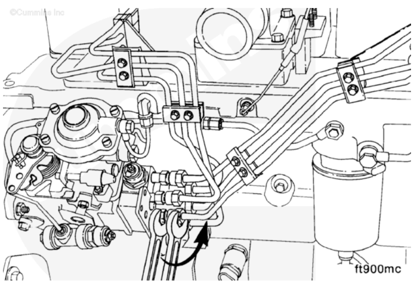
Disconnect the high-pressure fuel lines from the injectors, and complete the following steps:

Remove the fuel line clamp capscrews from the intake cover.



Remove the fuel lines from the fuel injection pump.

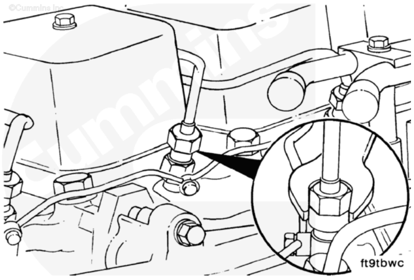
Note : Use two wrenches to prevent the delivery valve holder from turning.



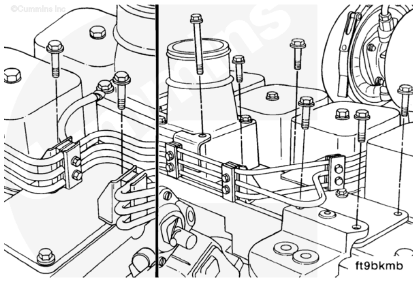
In-Line Pump

Note : If individual fuel lines are to be replaced, remove the support clamp from the set of fuel lines containing the line to be replaced.

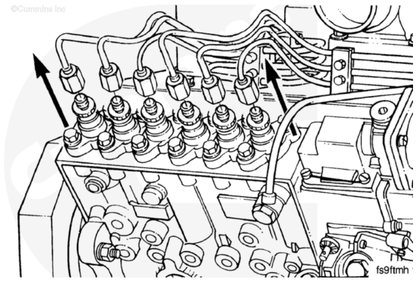
Disconnect the fuel line(s) from the injectors.



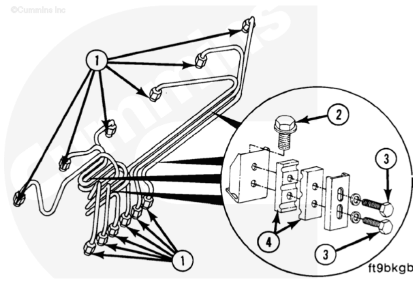
Remove the fuel line clamp capscrews from the intake cover.



Disconnect the fuel line(s) from the fuel injection pump.



Remove fuel line fittings (1), support bracket capscrews (2), vibration capscrew isolator (3), and isolators (4).



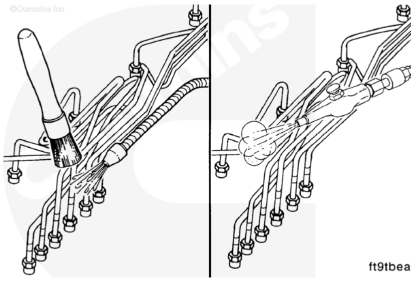
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

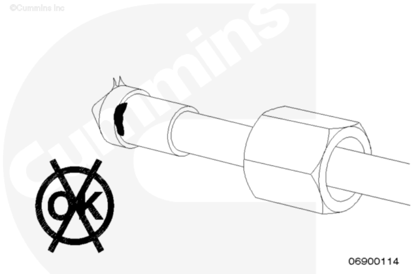


Note : Make sure that all paint chips are removed.

Wash the fuel lines in clean solvent.

Dry with compressed air.

Inspect the ferrules of the lines for any signs of burrs or foreign material.



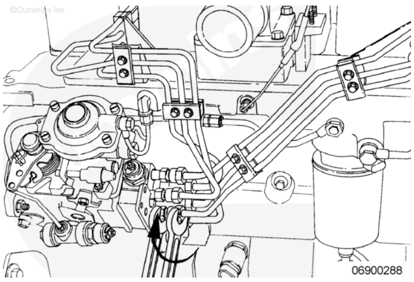
Install

Distributor-Type Pumps

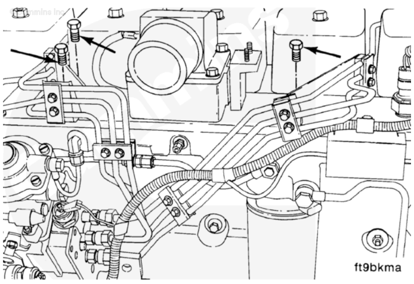
Install the fuel lines to the fuel injection pump.

Note : Use two wrenches to prevent the delivery valve holder from turning.

Torque Value: 24 n•m [18 ft-lb]

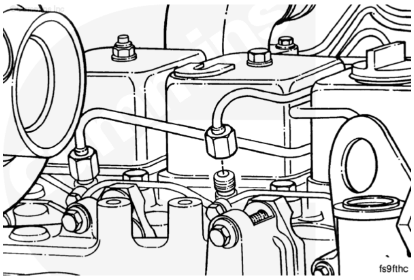


Install the fuel line clamp capscrews to the intake cover.



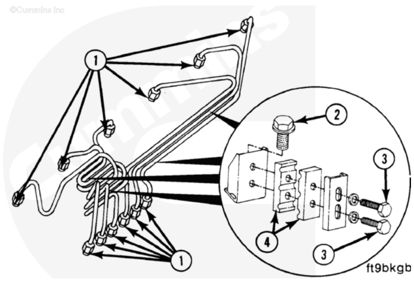
Connect the high-pressure fuel lines to the injectors.

Torque Value: 38 n•m [28 ft-lb]



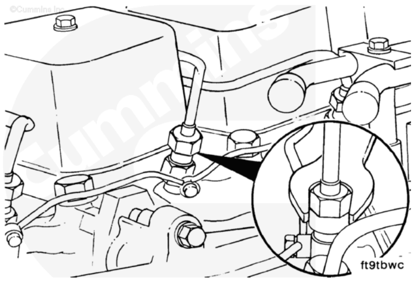
In-Line Pump

Install fuel line fittings (1), support bracket capscrews (2), vibration capscrew isolator (3), and isolators (4).



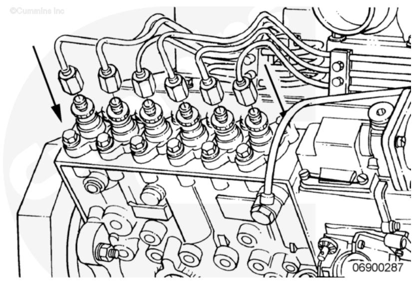
Connect the high-pressure fuel lines to the injectors.

Torque Value: 38 n•m [28 ft-lb]

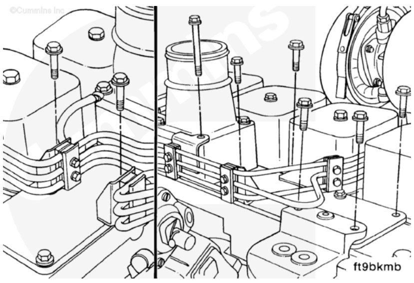


Connect the fuel line(s) to the fuel injection pump.

Torque Value: 24 n•m [18 ft-lb]



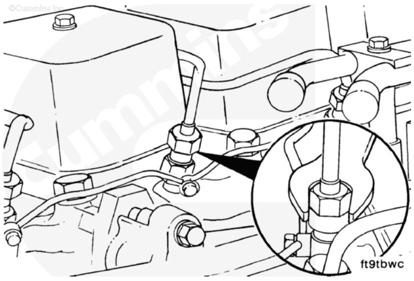
Install the fuel line clamp capscrews to the intake cover.



Connect the fuel line(s) to the injectors.

Torque Value: 38 n•m [28 ft-lb]

Note : **If individual fuel lines have been replaced, install the support clamp to the set of fuel lines containing the line that has been replaced.**



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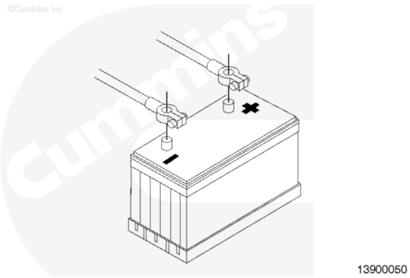
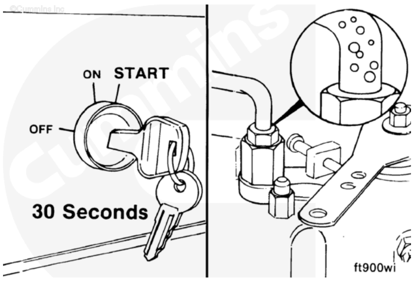


Connect the batteries.

Prime

High Pressure Fuel Line(s)

Replacing the fuel supply lines, fuel filters, fuel injection pump, high-pressure fuel lines, and injectors will let air enter into the fuel system. Follow the specified procedure to bleed the air from the system.

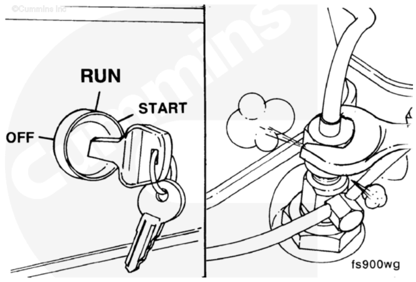


Refer to Procedure 006-015 (/qs3/pubsys2/xml/en/procedures/40/40-006-015-tr.html) Fuel Filter (Spin-On) for proper venting of the low pressure side of the fuel system.

Refer to the Procedure 005-012 (/qs3/pubsys2/xml/en/procedures/40/40-005-012-tr.html), Fuel Injection Pumps, In-Line, or Procedure 005-014 (/qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html), Fuel Injection Pumps, Rotary to determine if venting the fuel pump is necessary.

⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.



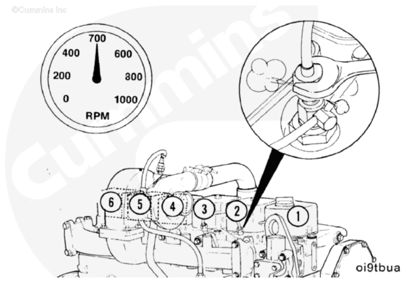
Check for air in the high-pressure lines by loosening the fittings at the cylinder head, and crank the engine to allow entrapped air to bleed from the line.

Tighten the fittings.

Torque Value: 38 n•m [28 ft-lb]

⚠ WARNING ⚠

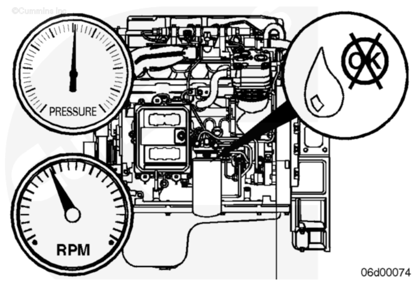
Do not vent the fuel system on a hot engine; this can cause fuel to spill onto a hot exhaust manifold, which can cause a fire.



Operate the engine, and vent one line at a time until the engine runs smoothly.

Finishing Steps

- Operate the engine and check for leaks.



Last Modified: 31-Mar-2006